

Project Report

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LOCALBIZ – AI-POWERED DIGITAL MARKETPLACE FOR LOCAL ENTREPRENEURS

1. Introduction

Small businesses, home-based entrepreneurs, artisans, and local producers play an important role in generating livelihoods and supporting local economic activity. However, many such entrepreneurs face challenges in reaching wider markets, establishing a digital presence, managing their businesses online, and effectively using digital technologies.

The proposed project, LocalBiz, is an AI-powered digital marketplace designed to help local entrepreneurs and small businesses establish an online presence and connect with customers beyond their immediate locality. The platform will provide features such as digital storefront creation, product listing, inventory management, order management, customer communication, digital payment support, and business analytics.

A key feature of the proposed platform is an AI Vendor Teaching Assistant, which will provide step-by-step guidance to entrepreneurs on how to use the platform and manage their digital business. This feature is particularly useful for users with limited digital experience.

The project is aligned with SDG 1 – No Poverty by supporting access to economic opportunities and livelihood generation, and with SDG 8 – Decent Work and Economic Growth by promoting entrepreneurship, productivity, and inclusive economic participation. The project also contributes to the broader vision of ViksitBharat 2047 by promoting grassroots entrepreneurship, digital inclusion, local economic development, and technology-enabled growth.

2. Problem Statement

Local entrepreneurs, artisans, and home-based businesses often face limited market reach, digital visibility, and access to simple online business tools. Many first-time entrepreneurs also lack the digital knowledge required to effectively manage an online business.

The proposed LocalBiz platform will provide a simple digital marketplace for product listing, inventory and order management, customer interaction, digital payments, and business analytics. An AI Vendor Teaching Assistant will provide step-by-step guidance to help entrepreneurs use the platform and develop digital business skills.

3. Aim of the Project

The aim of the project is to develop an AI-enabled digital marketplace that empowers local entrepreneurs, artisans, and small businesses to participate in the digital economy by providing accessible online selling, business management, and AI-based learning and guidance tools.

4. Objectives

1. To develop a user-friendly digital marketplace for local entrepreneurs, artisans, and small businesses.
2. To enable entrepreneurs to create digital storefronts and showcase their products to a wider customer base.
3. To provide simple tools for product listing, inventory management, order management, and customer communication.
4. To provide digital payment functionality through a secure test/sandbox payment environment.
5. To develop an analytics dashboard that helps entrepreneurs understand sales, orders, customer engagement, and product performance.
6. To develop an AI-powered Vendor Teaching Assistant that provides step-by-step guidance to entrepreneurs on using the platform and managing their digital businesses.
7. To promote digital literacy and technology adoption among entrepreneurs with limited experience in online commerce.
8. To improve market access and support income-generating opportunities for local entrepreneurs.
9. To promote entrepreneurship, productivity, and inclusive economic participation in alignment with SDG 8.
10. To contribute to SDG 1 and the Viksit Bharat @2047 vision through technology-enabled livelihood and entrepreneurship opportunities.

5. Proposed Solution

The proposed LocalBiz platform will act as a digital bridge between local entrepreneurs and customers.

Entrepreneurs will be able to:

- Register and create an online shop.
- Add and manage products.
- Upload product images.
- Set prices and stock quantities.
- Receive and manage customer orders.
- Communicate with customers.
- Track sales and business performance.
- Access AI-based guidance.

Customers will be able to:

- Register and log in.
- Browse local products.
- Search and filter products.
- View product details.
- Add products to a shopping cart.
- Complete checkout.
- Make test/sandbox payments.
- Track orders.

- Provide reviews and feedback.

Administrators will be able to:

- Manage users and vendors.
- Manage products and categories.
- Monitor orders and transactions.
- Monitor platform activity.
- View overall analytics.
- Track social and economic impact indicators.

6. Unique Feature – AI Vendor Teaching Assistant

The main differentiating feature of LocalBiz is the AI Vendor Teaching Assistant.

The purpose of this feature is to help entrepreneurs who have limited digital knowledge understand how to operate an online business.

The assistant will provide simple, step-by-step instructions for tasks such as:

- Creating an online shop.
- Adding a product.
- Writing product descriptions.
- Uploading product images.
- Managing inventory.
- Managing orders.
- Understanding digital payments.
- Understanding sales analytics.
- Improving product visibility.

Example

If a vendor asks:

“How do I add a product?”

The AI assistant will provide:

1. Open the Vendor Dashboard.
2. Select Add Product.
3. Enter the product name.
4. Enter the price.
5. Upload a product image.
6. Enter the available stock.
7. Add a product description.
8. Click Publish.

The AI assistant will therefore function not only as a chatbot but also as a digital learning and business-support tool for entrepreneurs.

7. AI Implementation

The AI Vendor Teaching Assistant can be implemented using a Retrieval-Augmented Generation (RAG) approach.

The system will contain a knowledge base consisting of:

- Platform user guides
- Vendor instructions
- Product management instructions
- Inventory instructions
- Order management instructions
- Payment instructions
- Analytics guides
- Frequently asked questions

AI Workflow

Vendor Question



Question Processing



Search Relevant Information



Retrieve Relevant Knowledge



AI/LLM Generates Response



Step-by-Step Guidance

This approach allows the assistant to provide answers based on the platform's approved knowledge rather than relying entirely on general AI knowledge.

8. Technology Stack and Tools

Component	Technology/Tool
Frontend	React.js
Styling	Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
Authentication	JWT, bcrypt
Image/File Storage	Cloudinary
Payment	Stripe/Razorpay Test API
Real-Time Communication	Socket.IO
Analytics	Recharts
AI Assistant	LLM API
AI Architecture	RAG
Vector Search	FAISS/ChromaDB
AI Backend	Python/FastAPI

Component	Technology/Tool
Document Processing	PyPDF
UI/UX Design	Figma
API Testing	Postman
Code Editor	VS Code
Version Control	Git, GitHub
Database Management	MongoDB Compass
System Diagrams	Draw.io
Deployment	Vercel, Render/Railway

9. System Modules

Module 1 – User Authentication

- Registration
- Login
- Logout
- Role-based access
- Customer/Vendor/Admin accounts

Module 2 – Vendor Management

- Vendor registration
- Digital shop creation
- Vendor profile
- Product management
- Inventory management

Module 3 – Product Management

- Add product
- Edit product
- Delete product
- Product images
- Categories
- Pricing
- Stock management

Module 4 – Customer Module

- Product browsing
- Search
- Filtering
- Cart
- Checkout
- Orders

- Reviews

Module 5 – Order Management

- Order creation
- Order status
- Vendor order management
- Customer order tracking

Module 6 – Payment Module

- Checkout
- Payment interface
- Test/sandbox transaction
- Payment status

Module 7 – AI Vendor Teaching Assistant

- AI chat interface
- Vendor guidance
- Step-by-step instructions
- FAQ support
- Platform learning assistance

Module 8 – Analytics

- Sales statistics
- Order statistics
- Product performance
- Customer engagement
- Vendor activity

Module 9 – Admin Dashboard

- User management
- Vendor management
- Product management
- Order monitoring
- Platform analytics
- Impact monitoring

10. SDG Alignment

SDG 1 – No Poverty

LocalBiz contributes to SDG 1 by improving access to digital markets and economic opportunities for local entrepreneurs and small businesses.

The project focuses on:

- Improved market access
- Digital inclusion
- Entrepreneurship
- Income-generating opportunities

- Access to digital business tools
- Support for economically vulnerable entrepreneurs

Impact Chain

Digital Inclusion → Market Access → Customer Reach → Business Opportunities → Livelihood Support

SDG 8 – Decent Work and Economic Growth

LocalBiz contributes to SDG 8 by promoting entrepreneurship, self-employment, productivity, and inclusive economic participation.

The project focuses on:

- Entrepreneurship development
- Digital business adoption
- Productivity improvement
- Small-business growth
- Economic participation
- Technology-enabled business opportunities

Impact Chain

Digital Entrepreneurship → Improved Productivity → Business Growth → Economic Participation → Inclusive Growth

12. Expected Outcomes

The expected outcomes of the project are:

- Increased digital participation of local entrepreneurs.
- Improved online visibility of local products.
- Wider customer reach.
- Simplified digital business management.
- Improved digital literacy through the AI teaching assistant.
- Better understanding of sales and customer engagement through analytics.
- Increased opportunities for local business growth.
- Promotion of entrepreneurship and inclusive economic participation.

13. Success Metrics

Platform Metrics

- Number of registered vendors.
- Number of registered customers.
- Number of digital shops created.
- Number of products listed.
- Number of orders processed.
- Number of active vendors.

Business Metrics

- Product views.
- Customer engagement.
- Number of transactions.
- Sales trends.
- Vendor activity.

AI Metrics

- Number of AI queries.
- Answer accuracy.
- Response time.
- Percentage of successfully resolved questions.
- Vendor satisfaction with AI guidance.

Social Impact Metrics

- Percentage of vendors successfully onboarded.
- Improvement in vendor digital confidence.
- Improvement in product visibility.
- Vendor feedback on market access.
- Change in business engagement before and after using the platform.

14. Conclusion

LocalBiz is proposed as an AI-powered digital marketplace designed to empower local entrepreneurs, artisans, home-based businesses, and small-scale producers. The platform combines digital commerce, business management tools, analytics, and an AI Vendor Teaching Assistant to reduce the barriers faced by entrepreneurs entering the digital economy.

The project goes beyond conventional e-commerce by focusing on entrepreneur empowerment, digital literacy, market access, and inclusive economic participation. Through its alignment with SDG 1 – No Poverty, SDG 8 – Decent Work and Economic Growth, and ViksitBharat2047, the project aims to demonstrate how technology can be used to support grassroots entrepreneurship and contribute to inclusive and sustainable economic development.